DISCUSSION

The purpose of this study was to correlate the high lipid per oxidation with histopathological alterations induced by sodium arsenite in the kidney of mice. Sodium arsenite contributes to the increased lipid per oxidation and altered histological appearances in the experimental mice. An earlier report suggests, Arsenic is an exogenously oxidative inducer, which can trigger the production of reactive oxygen species (ROS), causing damage to DNA, lipids and proteins.[14] Because the kidney is rich in phospholipids, and play a vital role in xenobiotic metabolism, is highly vulnerable to oxidative damage.[16]

In the present study, increased LPO levels in mice serum were found highly significant in sodium arsenite group (mean value 35.38, 37.12 and 44.01 nmol/ml for 4 weeks, 6 weeks and 8 weeks respectively) as compared to the control group (27.26 nmol/ml). Increased oxidative stress in tissue due to arsenic exposure is seemed to be the major cause of arsenic induced toxicity in mice. Santra *et al.* (2000)[25] has also reported oxidative stress in an experimental animal model due to arsenic exposure. Arsenic mediated oxidative stress can be indicated to induce the organ degeneration during the exposure. During the excretion process by the kidney, a significant amount of Arsenic inflows along with blood and other metabolites which further might interact with cellular protein or intercellular lipid or DNA molecules and thereby produce free radicals. This process may be responsible for the architectural changes of the kidney.[26] We have observed the degenerative changes, a cluster of nuclei, glomerulus diminished, the pervasiveness of cytoplasm; in kidney tissue in arsenic treated mice (figure 4-6). These damaging changes were observed as reflected action of toxic metabolites of Sodium arsenite. Microscopic changes were more pronounced in treated mice suffering from mild to high necrosis, indicating the toxic effect of chemical agents in the present study. Singh *et al.* (2011)[27] has also been reported renal failure, tubular necrosis due to arsenic toxicity. Apart from renal disorders, the toxic effect of arsenic has also been associated with hepatic disorders in an animal model study.[28]

It has been reported that exposure to arsenic is associated with metabolic disorders, hypertrophy of adrenal glands[29] and anaemia[30]. Exposure of sodium arsenite (2 mg/kg b.wt) to mice for the definite time intervals results in a significant increase of LPO levels in their serum.

CONCLUSION

Increased LPO leads to degeneration of lipids in the bilayer lipid layer of the plasma membrane which results in weak periplasm and hence nuclei inconsistently and irregularly present. Effect of LPO is applied in all the slides where irregularly spaced nuclei are observed i.e. 4th & 8th week treated slides.

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